



Julia Goh | juliagsy@gmail.com | Homepage: [Julia Goh](#) | LinkedIn: [Julia Goh](#)
| Referrals: [Petronas](#), [Unify](#), [UCL](#) | GitHub: [juliagsy](#) | Hugging Face: [juliagsy](#)

Experienced Machine Learning (ML) Engineer (Data Science, Generative AI and Software/Full-Stack). MEng Computer Science graduate at University College London. Fluent in English, Chinese and Malay. Open to roles in ML/AI Engineering and Data Science. Acquired Microsoft certifications of AI-900, AI-102 and DP-900.

Education

University College London (UCL) | MEng Computer Science | September '20 – June '24

- **Grade:** [First Class Honours](#)
- [Football Match Prediction](#): Regression, Classification, Tree and Forest, and SVM with **Python**.
- [Network Analysis of Programming Languages](#): Applied PageRank and HITS for analysing key driver in the evolution network, and community structure with modularity with **Python** and **NumPy**.

Experiences

Unify | Founding Engineer (AI/Full Stack) | April '25 – Present

- **LLM Interface**: Built the in-browser developer environment — terminal (streaming output, scrollbar, tab-completion), code editor with live execution, and a persistent CodeSandbox-backed file system with Google Drive sync.
- **AI Assistant Communications**: Architected the omni-channel communications infrastructure powering the AI assistant — voice calls, multi-channel messaging, emails, etc. — integrating Twilio, LiveKit and third-party APIs.
- [AI Desktop & Browser Control](#): Built a vision-based (VLM) computer-use agent that automates tasks across Ubuntu, Windows and macOS desktops, with installable clients for users to connect their own.
- **Organisation Support**: Built multi-tenant organisation features on top of personal workspaces — member management and invites, RBAC with custom roles, and project/resource sharing.
- **Integrations**: Designed a reusable OAuth integration framework and connectors with per-assistant connect/disconnect flows.
- **Tech Stack**: **Python, TypeScript, Node.js, React.js, Next.js, FastAPI, PostgreSQL, Docker, Kubernetes, GCP, Playwright, Git**

Petronas | Machine Learning Engineer (Intermediate) | September '24 – March '25

- **End-to-end model development and deployment**, such as data analysis, model training, hyperparameters tuning, model selection and cloud deployment on Azure.
- **Worked with linear models, trees, random forest, XG boost and light gradient boosting machine (LGBM)**.
- [PrAnCorr](#): ML. Built pipeline external corrosion depth & length prediction models of >90% accuracy.
- [PrAnSpan](#): ML. Built subsea pipeline height & length prediction models of >70% accuracy.
- **HIP: Computer Vision**. Worked on working person detection through object detection with YOLO, pose estimation and vLLM with GPT4o. Involved prompt engineering for better inference performance.
- **Tech Stack**: **Python, MLFlow, Docker, SQL, Azure, SonarCloud, DevOps, Git**

Unify | Machine Learning Engineer | June '22 – February '24

- [Ivy ML Models](#): *Applied Lead*. Led the team to expand ML model collection from scratch to 15+ within 4 months, including computer vision and NLP base models for business showcase.
- [Ivy Transpiler](#): Introduced the native compilation feature for optimising performance of ML models.
- [Model Hub](#): *Full stack*. Responsible in building client applications and API key authentication.
- [Console Application](#): *Cloud*. Involved in v0 development and API gateway authorisation.
- [Ivy](#): *Software Engineering*. Unified PyTorch, Tensorflow, JAX and NumPy functions, making them easily accessible through a single library. All codes are also readily compatible with the native libraries.
- **Tech Stack**: **Python, C++, CUDA, AMD RocM, Node.js, React.js, Next.js, TypeScript, Docker, GCP**

Research

UCL | Generative AI in Music | September '23 – April '24

- [Review](#). Studied existing tech related to image and music generations, majority being text-conditioned.
- [Paper](#). Developed POCs for Diffusion based Music-conditioned Image Generative AI and Transformer based Image-conditioned Music Generative AI.
- **Tech Stack: Python, Pillow, NumPy, SciPy, PyTorch, Transformers, Diffusers, Matplotlib, Git**

UCL | Multi-Agent Reinforcement Learning in Portfolio Management | January '23 – April '23

- [Paper](#). Explored the effect of different RL algorithms in competitive vs. cooperative MARL settings.
- Algorithms: Advantaged Actor-Critic (A2C), Deep Deterministic Policy Gradient (DDPG), Proximal Policy Optimisation (PPO)
- **Tech Stack: Python, OpenAI Gym, Stable Baselines 3, Pandas, Git**